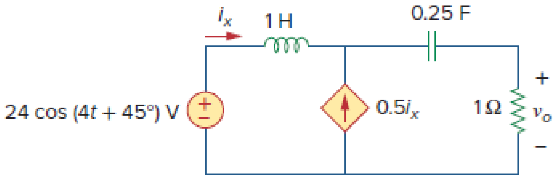
Chapter 10, Problem 4P

Problem

Compute *vo*(*t*) in the circuit of Fig. 10.53.

Figure 10.53:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in Figure 10.53 in the textbook.

Calculate the impedance value of inductor.

$$\begin{aligned} X_L &= j\omega L \\ &= j(4)(1) \\ &= j4\ \Omega \end{aligned}$$

Calculate the impedance value of capacitor.

$$\begin{aligned} X_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(4)(0.25)} \\ &= -j\ \Omega \end{aligned}$$

Step 2 of 3

The frequency domain equivalent circuit is shown in Figure 1.

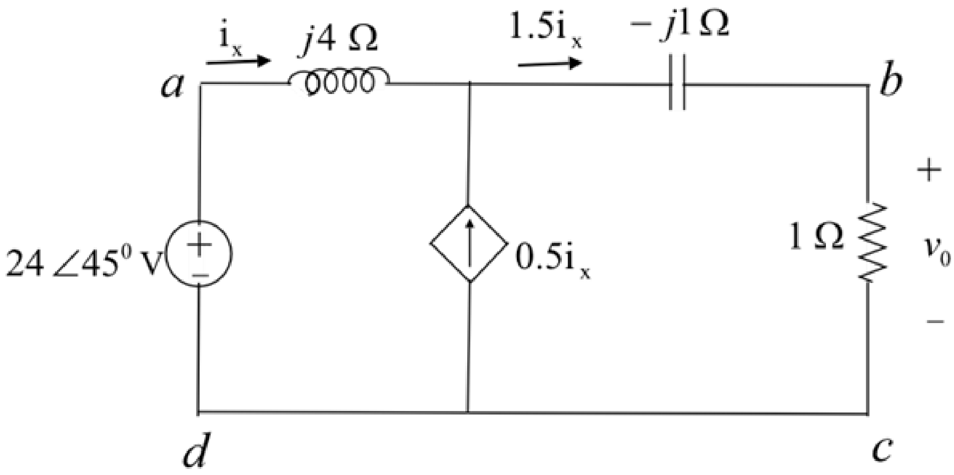


Figure 1

Step 3 of 3

Apply Kirchhoff's voltage law to outer loop of the circuit.

$$(j4)i_x + (-j+1)(1.5i_x) - 24\angle 45° = 0$$
$$(-j1.5 + 1.5 + j4)i_x = 24\angle 45°$$
$$(1.5 + j2.5)i_x = 24\angle 45°$$

$$\begin{aligned} i_x &= \frac{24\angle 45°}{2.915\angle 59°} \\ &= 8.23\angle -14° \text{ A} \end{aligned}$$

Calculate the output voltage using ohm's law.

$$v_o = (1\ \Omega)(1.5i_x)$$

Substitute $8.23\angle -14° \text{ A}$ for i_x.

$$\begin{aligned} v_o &= (1\ \Omega)(1.5)(8.23\angle -14° \text{ A}) \\ &= 12.3\angle -14° \text{ V} \end{aligned}$$

The output voltage in time domain form is,

$$v_o(t) = 12.3\cos(4t - 14°) \text{ V}$$

Therefore, the output voltage, $v_o(t)$ in the circuit is $\boxed{12.3\cos(4t - 14°) \text{ V}}$.